

Handout 1: The PPF and the Gains from Trade

Mankiw Ch 2-3

Answer key

Problem 1. The production possibilities frontier

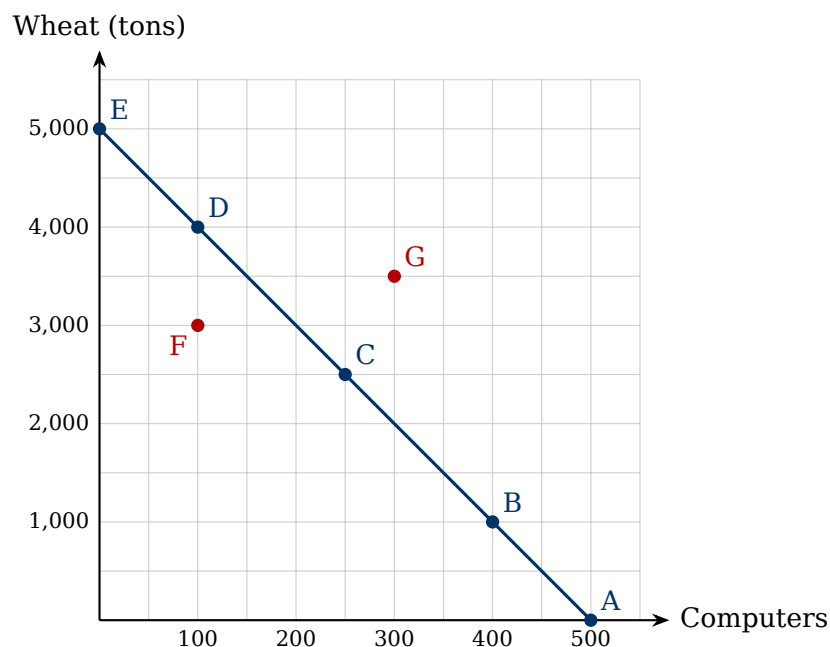
Two goods, computers and wheat. One input, labor: 50,000 hours a month. One computer takes 100 hours of labor; one ton of wheat takes 10 hours.

a) Complete the table.

Point	Labor hours		Output	
	Computers	Wheat	Computers	Wheat (tons)
A	50,000	0	500	0
B	40,000	10,000	400	1,000
C	25,000	25,000	250	2,500
D	10,000	40,000	100	4,000
E	0	50,000	0	5,000

Wheat hours = 50,000 – computer hours. Computers = hours/100; tons = hours/10.

b) Plot the five points and draw the PPF. Then locate F = (100 computers, 3,000 tons) and G = (300 computers, 3,500 tons). How many labor hours does each bundle need? Is each feasible? Efficient?



Hours needed = $100 \times \text{computers} + 10 \times \text{tons}$.

F: $10,000 + 30,000 = 40,000 < 50,000$. Feasible, but 10,000 hours are idle: inside the frontier, not efficient. The economy could have 100 more computers or 1,000 more tons for free.

G: $30,000 + 35,000 = 65,000 > 50,000$. Not feasible: outside the frontier.

- c) What is the opportunity cost of one computer? Of one ton of wheat? How does this show up in the graph?

One computer uses 100 hours, which would have grown $100/10 = 10$ tons of wheat: a computer costs 10 tons. One ton uses 10 hours = $1/10$ of a computer. The opportunity cost is the slope of the PPF: $5,000/500 = 10$ tons per computer, the same everywhere on the line because both goods use labor at fixed rates.

- d) Immigration adds 10,000 hours of labor a month. What happens to the PPF?

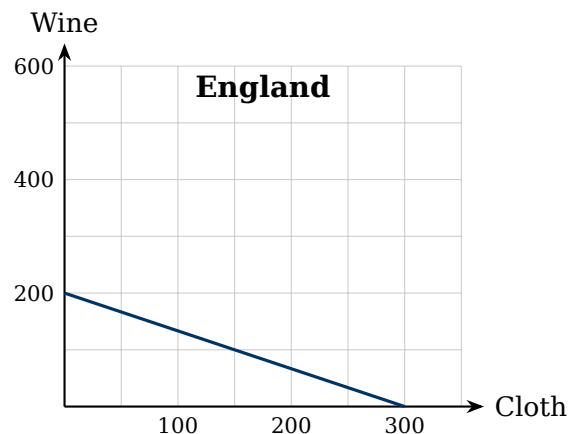
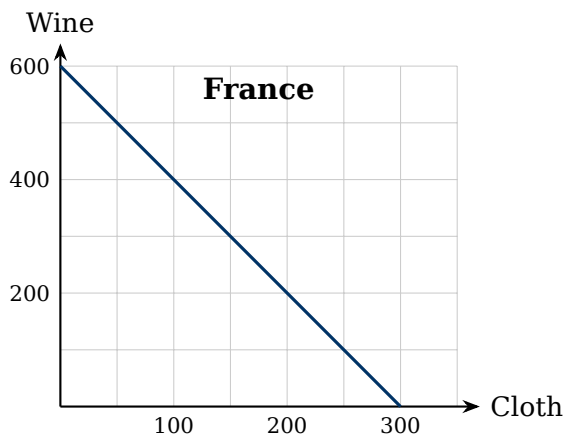
60,000 hours: 600 computers or 6,000 tons. Both intercepts rise by 20 percent, a parallel shift outward. The slope is unchanged: a computer still costs 10 tons.

- e) Instead, a new technology cuts the labor needed for a computer from 100 hours to 50. What happens to the PPF?

All labor on computers: $50,000/50 = 1,000$. All on wheat: still 5,000 tons. The frontier pivots outward around the wheat intercept. The slope changes: a computer now costs $50/10 = 5$ tons.

Problem 2. France and England

Each country's PPF for cloth and wine is drawn below.



- a) Calculate the opportunity cost of cloth and of wine in each country.

France: 300 cloth or 600 wine. One unit of cloth costs $600/300 = 2$ wine; one wine costs $1/2$ cloth.

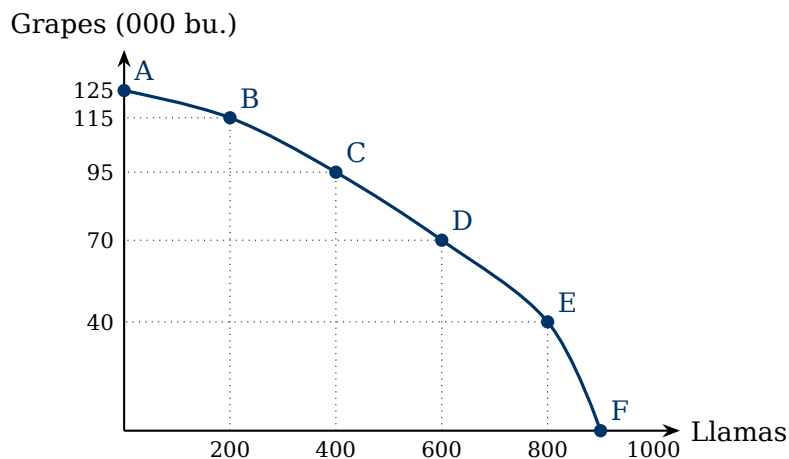
England: 300 cloth or 200 wine. One cloth costs $200/300 = 2/3$ wine; one wine costs $3/2$ cloth.

- b) Which country has the lower opportunity cost of cloth? Who should make what?

England: $2/3$ wine per cloth against France's 2. England has the comparative advantage in cloth, France in wine ($1/2$ cloth per wine against $3/2$). England makes cloth, France makes wine, and they trade at a price between $2/3$ and 2 wine per cloth.

Problem 3. Grapes and llamas

You bought land in California. Part of it is fertile and excellent for grapes; part is rocky and poor for grapes. Llamas do equally well on any part. Grapes are measured in thousands of bushels.



- a) You start at A, growing only grapes. What is the opportunity cost of raising the first 200 llamas? Of the next 200 (B to C)? Of the last 100 (E to F)?

A to B: grapes fall from 125 to 115, so 10 thousand bushels, or 50 bushels per llama. B to C: 20 thousand bushels, 100 per llama. E to F: 40 thousand bushels for 100 llamas, 400 per llama.

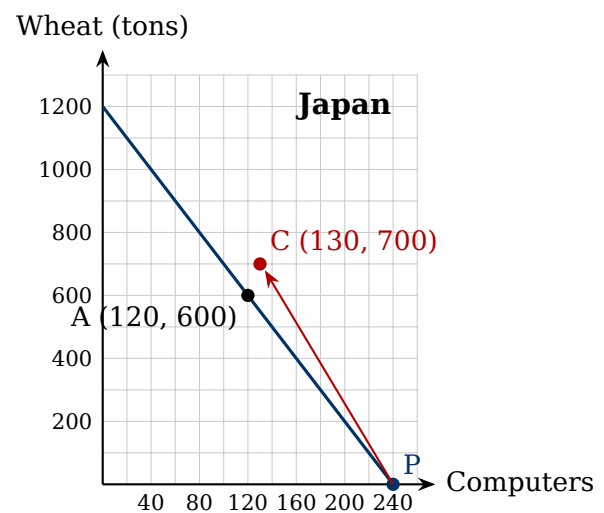
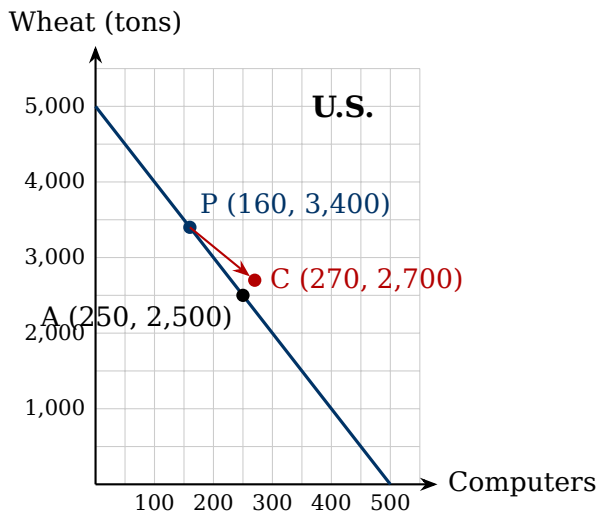
- b) The PPF is bowed outward. Explain why.

The first llamas go on the rocky land, which grew few grapes, so they cost little. As the herd grows, llamas displace grapes on better and better land, and each extra llama costs more grapes. Resources are not equally suited to both uses, so the opportunity cost of llamas rises as more are produced. A straight PPF (Problem 1) assumes identical resources and a constant opportunity cost.

Problem 4. The PPF and trade

Two countries, the U.S. and Japan; two goods, computers and wheat; one input, labor. The U.S. has 50,000 hours of labor; a computer takes 100 hours, a ton of wheat 10 hours. Japan has 30,000 hours; a computer takes 125 hours, a ton of wheat 25 hours.

a) Sketch each country's PPF.



U.S.: 500 computers or 5,000 tons. Japan: $30,000/125 = 240$ computers or $30,000/25 = 1,200$ tons. Both straight lines. Points A, P, and C from parts (b)-(e) are marked.

b) Each country puts half its labor into each good. Find production, which is also consumption, without trade.

U.S.: 25,000 hours each way: 250 computers and 2,500 tons (point A). Japan: 15,000 hours each way: $15,000/125 = 120$ computers and $15,000/25 = 600$ tons (point A).

c) The U.S. produces 3,400 tons of wheat. How many computers can it make with the rest of its labor? Mark the point on the PPF.

$3,400 \times 10 = 34,000$ hours on wheat; 16,000 hours left; $16,000/100 = 160$ computers. Point P = (160, 3,400).

d) Japan produces 240 computers. How much wheat can it make with the rest of its labor? Mark the point.

$240 \times 125 = 30,000$ hours, all of Japan's labor. Zero wheat. Point P = (240, 0).

e) Now the U.S. exports 700 tons of wheat to Japan and imports 110 computers from Japan. Find each country's consumption of both goods, compare it with part (b), and mark the consumption points.

U.S.: computers $160 + 110 = 270$; wheat $3,400 - 700 = 2,700$. Against (b): +20 computers and +200 tons.

Japan: computers $240 - 110 = 130$; wheat $0 + 700 = 700$. Against (b): +10 computers and +100 tons.

Both C points lie outside the country's own PPF: with trade each consumes a bundle it could not produce alone. World output rose from 370 to 400 computers

and from 3,100 to 3,400 tons with the same labor.

- f) Who has the absolute advantage in each good? The comparative advantage? Between what prices, in tons of wheat per computer, do both countries gain?

Absolute advantage: the U.S. in both (100 vs. 125 hours per computer, 10 vs. 25 per ton).

Opportunity cost of a computer: U.S. $100/10 = 10$ tons; Japan $125/25 = 5$ tons. Japan has the comparative advantage in computers, the U.S. in wheat (1/10 vs. 1/5 computer per ton).

Both gain if the price of a computer is between 5 and 10 tons. The deal in (e) is $700/110 \approx 6.4$ tons per computer.

Problem 5. Argentina and Brazil

Each country has 10,000 hours of labor. In Argentina a pound of coffee takes 2 hours and a bottle of wine 4 hours. In Brazil a pound of coffee takes 1 hour and a bottle of wine 5 hours.

- a) Which country has the absolute advantage in coffee? In wine?

Coffee: Brazil (1 hour vs. 2). Wine: Argentina (4 hours vs. 5).

- b) Which country has the comparative advantage in coffee? In wine?

Opportunity cost of a pound of coffee: Argentina $2/4 = 0.5$ bottle; Brazil $1/5 = 0.2$ bottle. Brazil has the comparative advantage in coffee.

Opportunity cost of a bottle of wine: Argentina $4/2 = 2$ lb; Brazil $5/1 = 5$ lb. Argentina has the comparative advantage in wine.

- c) Between what prices, in pounds of coffee per bottle of wine, would both countries gain from trade?

Between 2 and 5 pounds of coffee per bottle. Argentina sells wine for more than its cost of 2 lb; Brazil buys wine for less than its cost of 5 lb.